When Can a Bandit Trust Correlation? Certified Pooling under Persistent Web Rewards

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ABSTRACT

Web platforms hold many graphs, such as co-engagement, embeddings and hyperlinks, that seem to say which items share appeal. Their rewards are also restless: engagement persists for days and moves together across items. Bandits that pool information along such graphs, or that build confidence sets as if rewards were independent, can fail. We ask when a learner can trust correlation. We model each reward as a long-run mean, restricted by a graph certificate, plus a persistent autoregressive deviation whose shocks are correlated along a graph. For the mean channel, certified pooling gains in proportion to the components its certificate can reject. Across certificates, regret falls with the rejectable share (Spearman −0.87 on KuaiRec, −0.81 on the Open Bandit Dataset). Uncertified pooling fails on a minority of instances, and real graphs beat rewired ones only under certification. To bridge the two channels, we prove two results. Certified pooling stays valid under persistent, correlated, restless sampling when its confidence sets come from an exact innovation regression. Confidence sets built for independent rewards fail under bursty exposure, by an inflation factor we derive. Real Wikipedia attention is in this failure regime: a week of consecutive views carries the information of 1.3–1.9 independent days. On worlds calibrated to it, iid certificates are violated in 56–95% of runs, while innovation certificates hold but are still too wide to act on. On a chronologically valid KuaiRec panel, persistence-aware policies halve regret against Thompson sampling, but the last observed value captures most of the gain.

CCS CONCEPTS

• **Computing methodologies** → **Online learning settings**; • **Information systems** → *Recommender systems.*

KEYWORDS

multi-armed bandits, correlated arms, graph bandits, restless bandits, confidence sequences, recommendation

1 INTRODUCTION

Every day a short-video platform fills a trending slot. Two facts shape the decision. First, videos are related: those watched by the same audience tend to be enjoyed by it, so evidence about one video is evidence about its neighbours. Second, engagement persists: a video's completion rate today predicts tomorrow's, and shocks such as a holiday hit many videos at once. The platform also holds many graphs that might encode the first fact, such as co-engagement, matrix-factorization neighbours, creator links, tags and a social network. It must decide which of them to use, if any.

A learner can go wrong in two ways. It can *trust correlation that is not there*: pooling along a graph whose neighbourhoods do not share appeal biases estimates and can lead to linear regret. Or it can *ignore that rewards persist*: confidence sets built for independent rewards are too narrow when consecutive observations move together. Decisions that rely on them, such as dropping an item for good, then go wrong.

Relevance to the Web. Both mistakes are specific to Web platforms. Their graphs are by-products of logging (co-engagement, embeddings, hyperlinks, social ties), not priors designed for the decision at hand. Their rewards are restless, because attention and engagement persist and spread. Their exposure is bursty: the same items fill a slate day after day. Graph and cluster bandits are a staple of Web recommendation [2, 30, 32, 34], but they take the graph as given and the noise as independent. This paper asks what an online learner on a Web platform may trust, and how much each observation is worth. We answer with theory and with measurements on three Web platforms: KuaiRec short video [8], Wikipedia page views, and the Open Bandit Dataset [24].

Model and questions. We write every reward as $f_{i,t} = \mu_i + z_{i,t}$. In the *mean channel*, a graph certificate restricts the long-run appeal μ. In the *fluctuation channel*, the deviations z follow an autoregression with graph-correlated shocks. Graphs are a tool for encoding correlation in either channel, not the object of study. We ask three questions. How can a learner pool safely (§3)? What breaks when rewards persist (§4)? How should a learner act on persistent fluctuations (§5)? Each section states the theory, the prediction it makes, and what the data say.

Contributions.

(1) **A two-channel model of correlated arms** (§2). It contains iid, static graph and restless autoregressive bandits as special cases, and separates what a graph can encode.
(2) **Certified pooling and its data signature** (§3). The regret of certified pooling depends on which components the certificate makes *rejectable*. On two platforms the gain tracks the rejectable share across certificates. Uncertified pooling fails on a minority of instances, as the misalignment example predicts. Real graphs beat degree-preserving rewirings only under certification.
(3) **Theory bridging the channels** (§4). Certified pooling is valid under persistent, correlated, restless sampling when it is built on an exact innovation regression (B1), including a scalar construction for correlated shocks. iid certificates fail under bursty exposure by an inflation factor that we derive for any stationary autocovariance (B2). The mean-channel constants carry over with long-run covariances, and we give the general slate-contrast formula (B3).
(4) **Predictions tested on real dynamics.** B2's inflation appears in held-out Wikipedia attention, and the full autocovariance predicts it. iid certificates fail on worlds calibrated to it, while innovation certificates hold, including with estimated dynamics. On a chronologically valid KuaiRec panel, persistence-aware policies halve regret (§5).

What we do not claim. Graph structure adds little beyond shrinkage except through certified rejection. At Web persistence, today's anytime-valid certificates are valid but too wide to act on. On real panels a simple persistence rule matches the full joint filter. We report each of these limits with its evidence, because they are the useful practical guidance.

2 SETTING: TWO CHANNELS OF CORRELATION

There are N arms. In round t the learner pulls one arm, or a slate S_t of k arms, and observes

$$Y_t = \mu_{a_t} + z_{a_t,t} + \xi_t, \quad z_t = \sum_{r=1}^{p} A_r z_{t-r} + \eta_t, \quad \eta_t \sim N(0, Q_G), \quad (1)$$

with observation noise $\xi_t \sim N(0, r)$. Every arm evolves every round (restless), but only pulled arms are observed.

Mean channel. The long-run means $\mu \in [0,1]^N$ are unknown. A graph supplies a partition into components $\{C_c\}_{c=1}^{M}$ with certified widths $\varepsilon_c \geq \max_{i \in C_c} \mu_i - \min_{i \in C_c} \mu_i$, or an energy bound $\mu_c^\top L_c \mu_c \leq S_c^2$. An energy bound implies widths through effective resistance. Let $v_c = \max_{i \in C_c} \mu_i$ and let $\Gamma_c = \mu^\star - v_c$ be the component gap.

Fluctuation channel. Deviations persist through A_r and are correlated across arms through Q_G, which may be shrunk toward a graph kernel. A shock common to all arms (a holiday) and a shock common to a community (a league's game day) are both correlation, but only the second is something a graph can distinguish from a common factor.

Regret. *Mean regret* $\sum_t (\mu^\star - \mu_{a_t})$ measures how well the learner learns persistent appeal. *Dynamic regret* $\sum_t (\max_i f_{i,t} - f_{a_t,t})$ measures tracking against an oracle that sees every arm's current value. With slates, both are summed over the slate against the oracle's top k. Setting $A_r = 0$ and taking Q_G diagonal with no certificate gives iid bandits. Static graph bandits add a mean certificate. Spectral bandits [27] regularize the mean channel without certifying it.

3 MEAN CHANNEL: POOLING SAFELY

Certified pooling. SP-UCB keeps an arm upper bound $u_i = \bar{Y}_i + b(n_i)$ with $b(n) = \sigma\sqrt{2\ell/n}$. It also keeps a pooled estimate $\hat{p}_c$ over all N_c pulls in component c. It selects the component that maximizes

$$U_c = \min\{\max_{i \in C_c} u_i,\ \hat{p}_c + b(N_c) + \varepsilon_c\}$$

and pulls that component's best arm. SP-KLUCB replaces both radii with Bernoulli KL bounds [9] at the same fixed level.

Theorem 3.1 (A1: alignment-sensitive regret). *Let* $\ell = \log(2(N+M)T/\delta)$. *If the width certificates are valid, then with probability at least* $1-\delta$

$$\mathcal{R}_T \leq \sum_{c \in O} A_c + \sum_{c \notin O} \min\{A_c, H_c\},$$

where O *is the set of components containing an optimal arm,* $A_c = \sum_{i \in C_c : \Delta_i > 0} (\Delta_i + 8\sigma^2\ell/\Delta_i)$ *is the cost of exploring the component arm by arm, and* $H_c = (\Gamma_c + \varepsilon_c)(1 + 8\sigma^2\ell/(\Gamma_c - \varepsilon_c)^2)$ *is the cost of rejecting it as a whole.* H_c *is finite only when* $\varepsilon_c < \Gamma_c$, *that is, when the component is* rejectable.

Graph	Side	Quotient	Taste corr.	Choice cost
I-mf (factorization)	item	0.76 / 0.96	**0.046**	0.117 / 0.019
I-coeng (co-engagement)	item	1.10 / 1.30	**0.034**	0.131 / 0.037
I-cat, I-tag (content)	item	0.91 / 0.99	0.010	0.099 / 0.079
I-coauthor (creator)	item	0.81 / 1.01	0.004	0.040 / 0.184
U-coeng (co-engagement)	user	**0.65** / 0.79	**0.020**	0.284 / 0.291
U-mf (factorization)	user	0.85 / 0.91	0.016	0.300 / 0.298
U-geo, U-feat, U-demo	user	0.95–0.99	0.002–0.007	0.31–0.35

Table 1: Which KuaiRec graphs encode correlated appeal ($k = 10$). Quotient: smoothness of true means, real / rewired (random ≈ 1; lower is smoother). Taste corr.: correlation of neighbours' double-centred taste (random pairs: 0.000–0.002). Choice cost: users' loss in best-choice value from smoothing with $(I + 10L)^{-1}$, real / rewired.

Geometry matters beyond widths. A path and a clique with the same diameter and energy budget differ in information cost by a factor that grows with component size (Appendix A). Without a certificate, pooling is unsafe. Take $\mu = (1, 0.9, 0)$ with arms 1 and 3 linked: smoothing along the edge makes arm 2 look best, and a learner that trusts the smoothed values plays it forever. We call this failure *N1*. It requires an informative graph, because a random graph shrinks all arms toward a common value and preserves rankings.

Prediction 1. *(a) Certified pooling gains in proportion to the share of suboptimal components that its certificate makes rejectable. (b) Zero-width (uncertified) pooling fails when the optimal arm's component has a lower average than another component. (c) Average gains come mostly from shrinkage, so a real graph should beat its rewiring only through the components it makes rejectable.*

Which graphs carry taste. KuaiRec's 1,411 users × 3,327 videos matrix is 99.6% observed. A larger, disjoint log lets us build sixteen graphs without leakage (Appendix B; Table 1). Collaborative graphs carry personal taste. Factorization and co-engagement neighbours agree on double-centred taste at 17–23 times the random-pair level: 0.046 and 0.034 for item graphs against 0.002 for random pairs. Tag, category and creator graphs mostly group videos of similar popularity, and profile, location and demographic user graphs are close to random. Being smoother than chance does not make a graph safe to smooth. Fixed smoothing on the factorization graph costs users 0.117 of best-choice value, against 0.019 on its rewiring. This is N1's mechanism, visible before any bandit runs.

Design (Setting A).. For each of 120 held-out users and 3 video subsets (360 instances), we sample 300 videos and rebuild each item graph on them. SP-KLUCB uses a 20-component spectral partition, rewards are Bernoulli with the user's true means, $T = 20{,}000$, and all policies share common random numbers. We compare five certificates: oracle widths; energy (computed from the graph); population-calibrated quantiles over tuning users; per-user calibrated quantiles; and none. Ratios are to KL-UCB at the same fixed level, with a user bootstrap. Fixed-level KL policies lose to Thompson sampling in absolute terms (oracle SP-KLUCB is 1.46× TS), so every claim here is within the KL family.

Certificate (I-mf graph)	Regret ratio	90th pct.	Valid	Rejectable
Oracle widths	**0.64** [0.61, 0.67]	1.04	100%	59%
Per-user calibrated, q75	0.82	1.04	74%	–
Population calibrated, q50	0.70 [0.66, 0.73]	1.04	47%	62%
Population calibrated, q90	0.99 [0.98, 0.99]	1.04	90%	15%
Energy (from the graph)	0.96 [0.95, 0.97]	1.04	100%	21%
None (zero width)	0.37 [0.35, 0.40]	1.08	0%	95%

Table 2: The certificate gap on KuaiRec (Setting A, 360 instances). Regret ratio: ratio of mean regrets to KL-UCB with 95% CI. The 90th percentile is of per-instance ratios. Valid: share of components whose width covers the true range. Rejectable: share of suboptimal components with width below their gap.

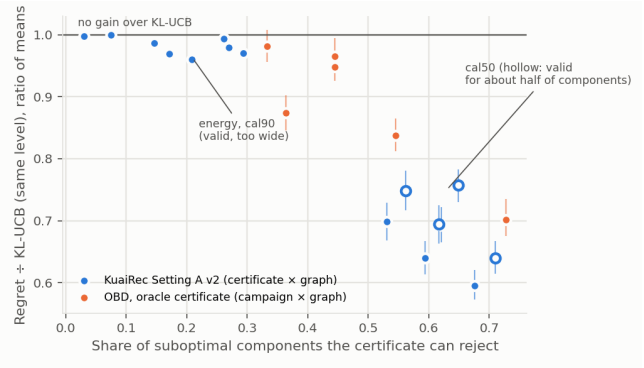


Figure 1: Certified pooling gains in proportion to the share of suboptimal components its certificate can reject. Each point is a certificate × graph cell on KuaiRec (360 instances) or a campaign × grouping cell on the Open Bandit Dataset (oracle certificate, 20 seeds). Hollow points are certificates valid for under 85% of components. Bars are 95% bootstrap intervals.

Findings on KuaiRec. Gains follow Theorem 3.1 across certificates (Prediction 1a; Table 2, Figure 1). Across the 16 certificate × graph cells, the regret ratio falls as the rejectable share rises (Spearman −0.87). Valid but wide certificates (energy, 90th-percentile calibration) reject 15–29% of suboptimal components and gain 1–4%. The oracle rejects 59–68% and gains 30–40%. Within a single certificate, the per-instance link is weak (Spearman −0.09 over 1,440 oracle instances). Rejectability therefore predicts *which certificate* helps better than *which instance* it helps.

Uncertified pooling fails on a minority of instances (Prediction 1b). Zero-width pooling is best on average (0.37) but carries the tail. On the tag graph its 90th percentile is 1.74× KL-UCB. Spectral Thompson sampling on the co-engagement graph has 1.54× TS's regret on average, and 27% of users exceed 2×. Failures occur when the optimal arm's component is not the top component by average, on informative graphs, and never on rewired graphs. Every certified variant stays at or below 1.04× at the 90th percentile.

Structure helps only under certification (Prediction 1c). Paired on the 360 instances, oracle-certified SP-KLUCB on the real factorization graph has 0.916× the regret it has on the rewired graph [0.905, 0.926], because the real graph makes more components rejectable (59% against 53%). Uncertified pooling (1.03 [0.96, 1.10]) and spectral Thompson sampling (1.00 [0.96, 1.05]) cannot be told apart from the rewired null. Graph-free shrinkage, Gaussian TS on the complete graph, is the best average policy (0.69× TS). When pooling across 300 users who arrive at random, shrinkage toward the population mean cuts regret by 27%, and no user graph beats it (Appendix C).

Replication on the Open Bandit Dataset. The ZOZOTOWN logs come from a uniform-random policy over 80 items (All campaign) and 46 items (Women campaign) [24]. We treat click rates on days 4–7 as true means. Components are built from days 1–3 only: attribute groups, audience-response clusters, and shuffled labels as the null. This is semi-synthetic evidence, with real parameters and simulated clicks. All three predictions replicate. The oracle gain tracks the rejectable share (Spearman −0.81 over six cells): the largest gain, 0.70, comes where 73% of components are rejectable, and gains are 2–5% where a third are. Uncertified pooling on audience clusters never finds the best item, so its regret grows linearly (959, 1,442 and 2,353 clicks lost at 0.5M, 1M and 2M impressions). Graph-free empirical-Bayes Thompson sampling is best by a wide margin (0.21–0.29× KL-UCB).

Answer. Pool only with a certificate, and expect gains in proportion to the components the certificate can reject. Benchmark against graph-free empirical-Bayes shrinkage, which delivers the average gain. The bottleneck is a certificate that is both valid and narrow. Per-user calibration recovers about half of the oracle's gain at 74% validity.

4 THE BRIDGE: CERTIFICATES UNDER PERSISTENCE

Certified pooling relies on confidence bounds. With persistent, restless rewards, consecutive observations of an arm are correlated, and an arm's current value is partly predictable from its past. The *innovation regression* removes the predictable part exactly.

Theorem 4.1 (Innovation regression). *Given a candidate mean μ, a Kalman recursion [16] that does not depend on μ gives, for the chosen arm, $R_t = u_t^\top \mu + \epsilon_t$, where u_t is predictable and $\epsilon_t \mid \mathcal{H}_t \sim N(0,1)$. Let $J_T = H + \sum_t u_t u_t^\top$ and $\hat{\mu}_T = J_T^{-1} \sum_t u_t R_t$, and let β_T be a self-normalized width [1]. Then, with probability at least $1-\delta$, $|v^\top(\hat{\mu}_T - \mu)| \le \beta_T \|v\|_{J_T^{-1}}$ for all T and all v.*

Theorem 4.2 (B1: certified pooling under persistence). *Build SP-UCB's arm and component bounds from Theorem 4.1, with component bound $\min_{w \in \Delta(C_c)} (w^\top \hat{\mu} + \beta_t \|w\|_{J_t^{-1}}) + \varepsilon_c$. If the certificates are valid and the dynamics correctly specified, every bound holds at every round with probability at least $1-\delta$, under any adaptive restless schedule. Suppose each pull adds at least ι information to the relevant contrast. Then Theorem 3.1 holds with $8\sigma^2\ell$ replaced by $4\beta_T^2/\iota$. For independent AR(1) arms, $\iota \ge (1-\varphi)^2/(V+r)$.*

φ	B2 sd inflation	iid violated	B1″ violated	iid drops best arm
0	1.00	0%	0%	–
0.5	1.68	0%	0%	–
0.9	3.49	85%	0%	28% (batch 25), 39% (100)
0.97	4.42	90–95%	0%	72% (batch 25), 89% (100)

Table 3: B2's prediction against observed failures, with blocks of 25 consecutive pulls. Violated: share of runs with any certificate violation. Drops: certified successive elimination, 100 worlds per cell, $T = 20{,}000$. Valid elimination never dropped the best arm. With one-at-a-time exposure, iid certificates were never violated and never dropped it.

The width β_T carries an N-dimensional log-determinant. Scalar constructions avoid it. If Q_G is diagonal, each arm and component statistic is a scalar martingale, and one-dimensional mixture bounds with a union bound over the $N+M$ statistics are valid (B1′). With correlated shocks, the other arms' means enter each arm's regression as bounded nuisances. Plugging in intervals that contain the truth yields intervals that contain the truth, so iterating from $[0, 1]$ is valid (B1″; Appendix A).

Proposition 4.3 (B2: iid certificates under persistence). *Suppose a stationary arm with autocorrelation $\rho(\cdot)$ is pulled in blocks of b consecutive rounds. Then the sample mean's variance is inflated by $f(b) = 1+2\sum_{h<b}(1-h/b)\rho(h)$, which for AR(1) is $\frac{1+\varphi}{1-\varphi} - \frac{2\varphi(1-\varphi^b)}{b(1-\varphi)^2}$. The iid radius $\sqrt{2\sigma^2\ell/n}$ therefore miscovers with per-check probability $2\Phi(-\sqrt{2\ell/f(b)})$.*

Proposition 4.4 (B3: long-run constants and slates). *(1) Under long runs or full slates with common AR coefficients, every mean-channel constant holds with σ^2 replaced by the long-run variance q/c^2, where $c = 1 - \sum_r a_r$. For AR(1) this is $V(1+\varphi)/(1-\varphi)$. (2) Two arms observed n times each have long-run contrast variance $(\Omega_{ii}+\Omega_{jj}-2\Omega_{ij})/n$, where Ω is the long-run covariance. With equal variances and common dynamics, this is the iid value times $1-\rho$.*

Prediction 2. *(a) iid certificates are violated under bursty exposure at rates that grow with persistence as B2 predicts, and they stay valid under interleaved exposure. (b) Innovation certificates are violated at most at rate δ. (c) Irrevocable decisions built on iid certificates inherit their failures. (d) Real Web attention shows the inflation that B2 and B3(1) predict from its fitted autocovariance.*

Controlled worlds. We use 20 arms in four components of five, with component widths 0.1 and best mean 0.55. Deviations are AR(1) with $\mathrm{Var}(z) = 1$ and $\varphi \in \{0, 0.5, 0.9, 0.97\}$, with independent shocks or shocks correlated along a path. Observation sd is 0.1 and $T = 5{,}000$. Exposure is either one arm at a time or blocks of 25 consecutive rounds, as in a daily slate or a batched update.

Findings in controlled worlds. B2 predicts the coverage failure (Prediction 2a; Table 3). One-at-a-time UCB interleaves arms, so repeat pulls are about 20 rounds apart and the inflation is small. The conservative union-bound level then still covers. *Innovation certificates are valid* (Prediction 2b): there were no violations in 2,080 runs with known dynamics, and later sweeps gave rates of 0–3%, consistent with $\delta = 0.05$. *Irrevocable decisions inherit the failure* (Prediction 2c). Once iid elimination drops the best arm, regret grows linearly at 0.14–0.26 per round. Validity has a price, though. At $\varphi \geq 0.9$, valid elimination eliminates nothing within 20,000 rounds, so at this horizon the invalid rule still has lower total regret. The scalar certificates cost little at low persistence: B1′ has 16% less regret than iid pooling at $\varphi = 0$.

Learner (B1″, blocks of 25, $\varphi = 0.97$)	Runs violated
Known dynamics	0/60
Fitted on a 365-round prefix	0/60
Fitted on a 100-round prefix	8/60 (13%)
same, persistence +2 standard errors	3/60 (5%)
Correlated truth fitted as independent	0%
Student-$t(3)$ shocks, known parameters	3–17%
AR(7) weekly-lag truth fitted as AR(1)	**23%**
iid certificate	88%

Table 4: Certificate validity with estimated and misspecified dynamics. An independent implementation with dense AR(20) truth finds 1/100 (known), 5/100 (365-day fit) and 40/100 (wrong order).

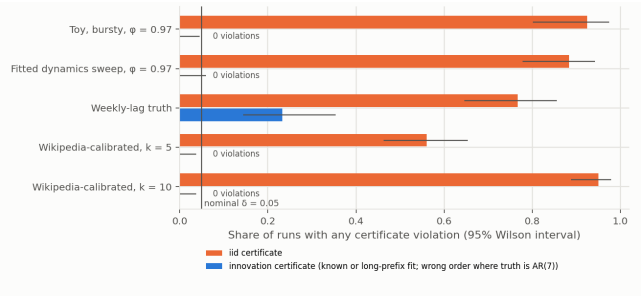


Figure 2: Share of runs with any certificate violation (95% Wilson intervals): the toy model, the estimated-dynamics sweep, a weekly-lag truth fitted as AR(1), and worlds calibrated to Wikipedia attention with slates of k articles.

Estimated dynamics. B1 assumes known dynamics, so we re-ran the bursty sweep with learners that fit persistence and shock correlation on a fully observed prefix (Figure 2). With a 365-round prefix there were no violations (0/60). A 100-round prefix biases persistence downward and gives 13% violations, or 5% after a two-standard-error margin. The dangerous misspecification is a *missed lag*. Fitting AR(1) when the truth is AR(7) with a weekly lag gives 23% violations under SP-UCB and 50–73% under elimination. Weekly structure is the norm in Web data (Table 4).

Real Web dynamics. For 300 Wikipedia articles, we fit the dynamics of $\log(1 + \text{views})$ on 2022–2023 and measure block-mean variance on 2024–2025 (Prediction 2d). After weekday adjustment, a 7-day block has 5.0–5.3 times the variance of seven independent days, and a 28-day block has 17–18 times. A week of consecutive exposure therefore carries the information of 1.3–1.9 independent days. The full autocovariance of an AR(20) fit predicts the held-out

Panel	Block	Observed	AR(1) shortcut	AR(20) full ACF
One domain	7 days	5.0	3.4	4.5
One domain	28 days	17.1	4.6	13.2
Six communities	7 days	5.3	4.4	5.0
Six communities	28 days	17.8	7.4	15.3

Table 5: Variance inflation of block means in Wikipedia attention: medians over 150 articles, weekday-adjusted, observed in 2024–2025 against predictions from 2022–2023 fits.

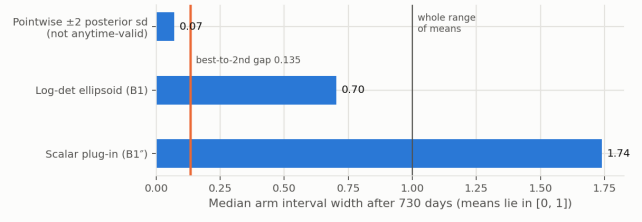


Figure 3: Median arm interval width after 730 days of 5-article slates on Wikipedia-calibrated dynamics with the true model. The information is there; the anytime-valid constructions do not exploit it.

inflation within 4–23% (median). The lag-1 shortcut underpredicts by a factor of 1.2–3.3, so certificates must use the full autocovariance (Table 5).

We then simulated 730-day worlds from AR(7) dynamics and an innovation covariance fitted to 30 articles in six communities, with resampled real residuals. The learner sees a slate of 5 or 10 articles a day. iid certificates are violated in 56% and 95% of runs. Innovation certificates are violated in 0–1%, with known dynamics, re-fitted AR(7) dynamics and even AR(1)-fitted dynamics. *But they are too wide to act on.* Valid elimination eliminates nothing in 730 days, and valid SP-UCB has about ten times the regret of the iid rule. Figure 3 shows why. After two years the posterior is informative: the pointwise width is 0.07, inside the 0.135 best-to-second gap. The log-determinant ellipsoid inflates it to 0.70. The B1″ plug-in reaches 1.74, because shocks correlated across 30 arms couple every interval to every other. The theorem is not at fault here. What is missing is a per-contrast anytime bound that pays $\sqrt{\log\log n}$ [14] rather than $\sqrt{N \log T}$ after whitening the shocks.

Slates. We checked B3(2) exactly, and Monte Carlo agrees within 0.3%. The factor $1 - \rho$ is exact for pairs with equal variances and dynamics, and the general long-run form is exact throughout. Under persistence the gain does not require simultaneous observation: staggered observations keep 96–99% of it, whereas under iid rewards they keep none.

Answer. Persistence must enter confidence sets. Under bursty exposure, iid certificates are violated most of the time, by the amount B2 predicts, and real Web attention is in that regime. Innovation certificates stay valid, including with dynamics estimated from a long prefix with the right lags. At Web persistence, however, current anytime-valid constructions are 10–25 times wider than the information allows.

Policy (regret ÷ iid Thompson sampling)	$k = 1$	$k = 5$
Joint filter, greedy	**0.47** [0.25, 0.70]	0.44 [0.35, 0.52]
Last observed value	0.49 [0.28, 0.74]	**0.43** [0.31, 0.55]
Temporal-only filter, greedy	0.50	0.46
Joint filter, UCB	0.51 [0.30, 0.73]	0.46 [0.38, 0.54]
rewired graph / common factor only	0.50 / 0.50	0.47 / 0.48
EWMA (tuned half-life)	0.49	0.73
Sliding-window UCB	0.61	0.74
Discounted Thompson sampling	0.77	0.90
Static (fit-window mean)	1.01	1.06

Table 6: KuaiRec daily slot under a chronological protocol: regret per slot per day relative to iid Thompson sampling, over 15 panel-window units, with bootstrap intervals over windows and then panels.

5 FLUCTUATION CHANNEL: ACTING ON PERSISTENCE

Stacking long-run means and AR lags gives a single Gaussian state, and every observation updates it for every arm and lag through a joint filter. Every causal policy has dynamic regret at least $T\{\mathcal{G}(\mu, K_0) - \mathcal{G}(\mu, K_0 - Q_G)\}$, where $\mathcal{G}(\mu, K) = \mathbb{E}\max_i(\mu_i + Z_i)$ with $Z \sim N(0, K)$ and K_0 is the stationary covariance. This is the *innovation lower bound.* Current-state Thompson sampling draws the fresh innovation, which no future observation will reveal, so exploring it is pure cost. Predictive sampling [23] draws only the part that the next round's rewards would reveal. We call this principle P1: explore only uncertainty that persists.

Prediction 3. *(a) Persistence-aware policies beat iid policies on persistent data. (b) Current-state TS underperforms policies that explore persistent uncertainty. (c) A graph-shrunk shock covariance helps when several items are compared at once.*

Controlled benchmark. We use 100 arms on a grid with AR(20) dynamics (coefficients summing to 0.97), shocks correlated along the grid, and $T = 2{,}000$. Joint predictive sampling, which has no tuning parameter, cuts dynamic regret by 29% against TS and closes 37% of the gap to the innovation lower bound. Current-state TS gains only 5% over TS (Prediction 3b). Correlated shocks add 15% on top of persistence, and the gain vanishes with independent shocks (Appendix C).

KuaiRec daily slot, chronologically. The panel has 63 days of daily complete-play rates. At each of three cutoffs, the co-engagement graph is rebuilt from interactions dated before the first test day. Eligibility is known at the cutoff, and a chosen video without plays returns nothing. There are three disjoint 11-day test windows, each crossed with five predefined panels of 49–50 videos (15 units), and budgets of $k = 1$ and 5 videos a day. Every hyperparameter is tuned on a development window and then frozen.

Findings. Persistence pays (Prediction 3a; Table 6). The joint filter, greedy or UCB, cuts regret by 49–53% ($k = 1$) and 54–56% ($k = 5$) against iid TS, and all intervals exclude zero. Standard nonstationary bandits [10], discounted TS and sliding-window UCB, are clearly worse. *The simplest persistence rule captures the gain.* Against the last observed value, the joint filter is inconclusive: +3%

at $k = 1$ and +7% at $k = 5$, with intervals spanning zero. *Shared shocks help when several items are compared each day* (Prediction 3c, Proposition 4.4(2)). At $k = 5$ the shock covariance cuts regret by 11% [5%, 20%] against the temporal-only filter, and at $k = 1$ it does nothing. Most of that gain comes from a common factor: the prefix graph cannot be distinguished from its rewiring or from a common factor alone. Retrospective Wikipedia panels give the same direction (Appendix C). On the Open Bandit Dataset, click-rate residuals do not persist (lag-1 autocorrelation -0.14 to -0.22), so there is no fluctuation channel to exploit.

Answer. Account for persistence: it is the largest real-data gain. On real panels, the last observed value captures most of it, so the case for a joint model rests on its uncertainty (certificates and exploration targeting), not on its point forecasts. Put shared shocks into the model when several items are compared at once. A common factor captures most of their value.

6 RELATED WORK

Graph, cluster and correlated-arm bandits. Spectral bandits assume means smooth on a graph and bound regret by an effective dimension [18, 27]. Graph-based pure exploration makes the information cost of similarity explicit [25]. Clustering approaches pool across users or arms [2, 4, 11], and social and knowledge graphs couple bandit parameters in Web recommendation [30, 32, 35]. Shared key-terms do the same in conversational recommendation [22, 34]. Correlated-arm UCB pulls non-competitive arms $O(1)$ times when correlation bounds are known [13], and network spillover creates dependence across users [6]. These works take the structure as given. We certify it, show that its value lies in rejectable components, and measure when Web graphs earn trust. Restricting exploration to uncertain comparisons echoes online pairwise ranking [15]. Our finding that shrinkage carries the average gain agrees with empirical-Bayes and prior-based Web systems [29, 33].

Non-stationary, autoregressive and restless bandits. Switching and drifting bandits forget old data [3, 10], and Web recommenders handle changing interests with model ensembles [31]. Closer to our fluctuation channel, rewards have been modelled as AR processes [5], latent AR states [26] and linear Gaussian dynamical systems with Kalman prediction [12]. Predictive sampling targets information that persists [23], and the knowledge gradient handles correlated beliefs [7]. We credit these works for filtering and for the policies. Our contribution is the bridge: the validity of certified pooling under persistence (B1), the failure of iid confidence (B2), and long-run constants (B3), each tested on real dynamics.

Confidence and evaluation on Web logs. Self-normalized bounds [1] and time-uniform confidence sequences [14] underlie our certificates. Recent work finds off-policy conclusions on the Open Bandit Dataset fragile, partly because its logs violate i.i.d. assumptions [19], which matches B2. Offline-online combinations [21] and slate evaluation [17] address related uses of logged Web data. Audits of short-video feeds show exploration at scale [28], the setting that LinUCB's news experiments opened [20].

7 CONCLUSION AND LIMITATIONS

Web correlation is valuable but has to be earned. For long-run appeal, collaborative graphs carry taste, but they help beyond shrinkage only through the components a certificate can reject. Uncertified pooling wins on average and fails on a minority of instances. For persistence, every confidence statement must account for it: real attention is in the regime where iid certificates fail, and innovation certificates repair validity. For acting, persistence roughly halves regret on a chronologically valid panel, and a simple rule captures most of the gain. We give practitioners five rules: compare each graph with its rewiring before trusting it; certify pooling and benchmark it against empirical-Bayes shrinkage; fit the full lag structure, including weekly lags; use persistence first; and explore only uncertainty that persists.

Limitations. KuaiRec and Open Bandit Dataset ground truths are estimated, so the mean-channel results are model-based evidence on real parameters, not logged-feedback evidence. The panel experiments assume that showing an item does not change its organic demand. The Wikipedia shock panels are retrospective. Replication units are modest: 120 users, 15 panel-window units, and 20 seeds per fixed instance. The main open problem is a tighter anytime-valid certificate under correlated persistent shocks. The information exists, but current constructions are 10–25 times wider.

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A PROOFS

Theorem 3.1. Fix an arm or component. Let Z_t be its selected-noise sum and N_t its count. Then $\exp(uZ_t - u^2\sigma^2 N_t/2)$ is a supermartingale. Stopping it at the nth pull and taking $u = b(n)/\sigma^2$ bounds each tail by $e^{-\ell}$. A union bound over $N+M$ processes and $n \le T$ gives an event $\mathcal{E}$ of probability at least $1-\delta$. On $\mathcal{E}$, the pooled mean B_{c,N_c} of the pulled arms satisfies $v_c - \varepsilon_c \le B_{c,N_c} \le v_c$, so both terms of U_c bound v_c. Optimism gives $\mu^\star \le U_c \le u_i \le \mu_i + 2b(n_i)$, so $n_i(T) \le 1 + 8\sigma^2\ell/\Delta_i^2$, which yields A_c. The pooled term gives $\mu^\star \le v_c + 2b(N_c) + \varepsilon_c$. Hence, if $\Gamma_c > \varepsilon_c$, then $N_c(T) \le 1 + 8\sigma^2\ell/(\Gamma_c - \varepsilon_c)^2$, and each such pull costs at most $\Gamma_c + \varepsilon_c$, which yields H_c. Since H_c increases in ε_c, tightening a valid certificate never worsens the bound. At zero width, the logarithmic terms of A_c and H_c differ by the factor $|C_c|$.

Path against clique. Take components with equal diameters and energy budgets. Endpoint observations reject a path component through its resistance geometry, while a clique requires arm-wise exploration, so the information constants differ by a factor that grows with $|C_c|$. The full statement and proof are in the supplementary material.

Theorem 4.1. Conditional on μ, the Kalman filter for z gives a one-step predictive mean that is affine in μ with predictable coefficients, and a predictive variance that does not depend on μ. Standardizing gives $R_t = u_t^\top \mu + \epsilon_t$ with $\epsilon_t \mid \mathcal{H}_t \sim N(0,1)$. The claim then follows from the self-normalized bound for vector martingales [1].

Theorem 4.2. Define $U_i = \hat{\mu}_i + \beta_t \|e_i\|_{J_t^{-1}}$ and the component bound in the statement. *Validity:* Theorem 4.1 holds for all directions at once, hence for every e_i and every w in every simplex. For $w \in \Delta(C_c)$, $w^\top \mu \ge v_c - \varepsilon_c$. Minimizing over w is free because the event does not depend on w. *Regret:* if c is selected at round t, then $\mu^\star \le U_c \le v_c + \varepsilon_c + 2\beta_t \|w\|_{J_t^{-1}}$. Hence $\min_w \|w\|^2_{J_t^{-1}} \ge (\Gamma_c - \varepsilon_c)^2/(4\beta_t^2)$, and the information-growth condition makes this at most $1/(\iota(N_c - 1))$. The rest of the summation follows the proof of Theorem 3.1. *AR(1) constant:* with independent arms the information is diagonal. In a pull of arm i, the coefficient of μ_i is $1 - \varphi^h s$, where $h \ge 1$ is the time since the arm's last observation and $s \in [0,1]$. This coefficient is at least $1 - \varphi$, and the predictive variance is at most $V + r$.

B1′ and B1″. With Q_G diagonal and $H = \alpha I$, each arm's score $W_i = \sum_{t:a_t=i} u_t \epsilon_t$ is a scalar martingale. The method of mixtures then gives $|W_i| \le \sqrt{2J_{ii}\log(\sqrt{J_{ii}/\alpha}/\delta')}$ uniformly in time, and a union bound over $N+M$ statistics completes B1′. With correlated shocks, let $G = J - \alpha I$. Then

$$q_i - \sum_{j \ne i} G_{ij} m_j = G_{ii}\mu_i + \sum_{j \ne i} G_{ij}(\mu_j - m_j) + W_i.$$

Given intervals that contain the truth, with midpoints m_j and half-widths ρ_j, the error of $\hat{\mu}_i$ is at most $(\sqrt{2J_{ii}\log(\sqrt{J_{ii}/\alpha}/\delta')} + \alpha + \sum_{j \ne i} |G_{ij}|\rho_j)/J_{ii}$. By induction from $[0,1]$, every interval contains the truth on the event that all scalar bounds hold. For components, write $\mu_i = \theta_c + d_i$ with $|d_i| \le \varepsilon_c/2$. The same argument then applies to θ, with a deterministic deviation term. The cross terms

$\sum_j |G_{ij}|\rho_j$ are what make the bound wide when many arms share correlated shocks (Figure 3).

AR(p). Stack the lags into a companion state. The augmented filter tracks $s_t \mid \mathcal{H}_t \sim N(g_t + D_t\mu, P_t)$. Observing arm a gives the design vector $u = e_a + D_{a\cdot}$, and (g, D, P) is updated by the Kalman gain. Theorem 4.1 applies unchanged, and with $p = 1$ this reproduces the AR(1) recursion.

Proposition 4.3. Within a block, the autocovariances are $V\rho(|s-u|)$, and summing gives $\mathrm{Var}(\sum) = Vb\,f(b)$. Blocks separated by other arms' blocks are nearly independent, so the sample mean is Gaussian with variance $(V/n)f(b)$. This gives the stated miscoverage. For AR(1), $\rho(h) = \varphi^h$ gives the closed form.

Proposition 4.4. (1) After the initial lags, the full-field innovation transform $X_t - \sum_k \mathrm{diag}(a_k)X_{t-k} = c\mu + \eta_t$ adds information $c^2 Q_G^{-1}$ per field. Substituting this for count-based information in the change-of-measure argument replaces σ^2 with q/c^2. (2) The contrast sum has variance $\sum_{s,u}[\gamma_{ii} + \gamma_{jj} - 2\gamma_{ij}](s-u)$. Dividing by n^2, it converges to $(\Omega_{ii} + \Omega_{jj} - 2\Omega_{ij})/n$. For staggered observations, the lag-one cross-covariances replace the lag-zero ones. Under persistence they are nearly equal, and under iid rewards the lag-one terms vanish.

B DATA AND GRAPHS

KuaiRec [8]. The small matrix has 1,411 users × 3,327 videos and 4.68M interactions. The big matrix has 7,176 × 10,728 and 12.5M interactions, with no overlap with the small matrix. The reward is min(watch ratio, 5)/5. Graphs are union k-NN graphs ($k \in \{5, 10, 20\}$), with Laplacians scaled to mean degree one. Nulls are degree-preserving rewirings. User-side signals use only non-evaluation videos, and video-side signals use only non-evaluation users. Users are split 50/50 into tuning and test sets.

Wikipedia. We use daily user page views for 2022–2025 from the public Wikimedia REST API, and hyperlinks from a 2026 snapshot. The one-domain panel has the 150 most-viewed programming-language articles. The six-community panel has 25 articles each from NFL, NBA and MLB teams, Premier League clubs, UN member states and programming languages.

Open Bandit Dataset [24]. The data are ZOZOTOWN uniform-random logs from 24–30 November 2019: the All campaign (1.37M impressions over 80 items) and the Women campaign (0.86M impressions over 46 items).

C ADDITIONAL RESULTS

Setting B (user graphs). We pool across 300 users who arrive at random, with 100 videos and T = 100,000. Shrinkage toward the population mean cuts regret by 27% against per-user TS. User-side pooling without a certificate is 1.39× per-user TS on all 8 instances. Real user graphs perform like their rewirings (0.99–1.02).

Wikipedia attention (retrospective). The task has ten featured slots a day, with models fitted on a rolling 365-day window. The joint filter cuts regret by 32% (one domain) and 24% (six communities) against iid TS. On the six-community panel, hyperlinks carry community shocks: the correlation along edges is 0.273, against 0.051 for rewired edges. They still change regret by at most ±3%, because long-run levels decide which articles lead. An independent chronological study on two other panels found the result panel-dependent.

Policy (dynamic regret per round)	Main	Indep. shocks	Weak persist.
ST UCB, 2 sd (tuned)	**0.850**	**0.890**	1.856
ST predictive sampling	0.950	1.057	1.568
SpectralUCB	1.026	1.014	1.953
UCB1	1.095	1.157	1.867
AR predictive sampling	1.121	1.094	**1.525**
ST current-state TS	1.258	1.452	2.246
Thompson sampling	1.329	1.523	1.675
Innovation lower bound	0.298	0.323	1.276

Table 7: Spatiotemporal benchmark: 100 arms, AR(20), T = 2,000**, 8 seeds.**